

AI Forward» Deployment Engineering (FDE)

/// End-to-End Solution Deployment ///





A complete zero-to-FDE program. You keep the full technical foundation (Python, ML, Deep Learning, Transformers, GenAI, Agentic AI, Cloud, DevOps) – but the real focus is the FDE job itself: understanding the client, translating the problem, building fast, and shipping a live solution.

The FDE reality this course is built around: 20% AI Expert · 30% Engineer · 50% Problem-Solver + Communicator + Shipper Foundations are the on-ramp. Discovery, translation, delivery, and deployment are the destination and they run through the entire course, not just one module.

The FDE who wins is not the one who knows the most about transformers. It's the one who can walk into a room, understand the client, build the right thing fast, and ship it.

What is a Forward Deployed Engineer (FDE)?

A Forward Deployed Engineer (FDE) is a hybrid engineer + consultant who works directly with a client to build and ship a working solution inside the client's real environment — not a generic product in a lab.



An FDE sits at the intersection of three worlds:



Engineering

They write real code, integrate systems, deploy to production, and debug live.



Consulting

They talk to stakeholders, translate messy business problems into technical requirements, and manage expectations.



AI

They know how to apply modern AI (LLMs, RAG, agents) to solve those problems fast.



The core difference:

A regular Software Engineer builds a **product for many users**. An FDE builds a **solution for one client**, tuned to that client's data, constraints, and goals.

The FDE balance — what this course is weighted around:



20% AI Expert

Enough AI depth to pick and apply the right approach.



30% Engineer

Build, integrate, deploy, and keep it running.



50%

Problem-Solver + Communicator + Shipper

Understand the client, translate the problem, present the solution, & deliver it live.

Course Structure - 5 Pillars

This is the full course content, organized Module 1 → Module 22 in sequence, grouped under 5 pillars - followed by 5 hands-on end-to-end projects:



1 Data and Programming



2 Software Engineering



3 AI and Analytics



4 Deployment Engineering



5 Client Delivery & Business Consulting

PART 1 - DATA AND PROGRAMMING



Module 1 – Python & Programming Foundations (7 lessons)

- **1.1 Python Essentials** | Variables, data types, control flow, functions, OOP basics
- **1.2 Working with Data in Python** | Lists, dicts, files, JSON, pandas & numpy basics
- **1.3 Python for AI** | requests, virtual environments, pip, async basics
- **1.4 Git & GitHub** | Version control, branches, commits, pull requests, collaboration
- **1.5 Linux & Command Line Basics** | Shell, files, permissions, environment variables
- **1.6 Clean Code & Debugging** | Readable code, error handling, logging, testing basics
- **1.7 DSA for AI Systems (High-Level)**



- Time & space complexity (Big-O) — how to reason about it
- Why an inefficient pipeline breaks at scale
- Core data structures and where each shows up in AI systems: arrays, hash maps, trees, graphs, heaps/priority queues
- Search & sort algorithm basics
- Picking the right data structure for fast lookups, retrieval & agent memory
- Recursion & basic dynamic programming — where these patterns quietly show up in AI code
- Why vector search (used in RAG) is really just a nearest-neighbor search problem at its core



Module 2 – Databases for FDEs (6 lessons)

- **2.1 Databases for FDEs |** SQL vs NoSQL; CRUD; connecting a database to your app
- **2.2 Relational Databases Deep Dive |** Tables, schemas, primary/foreign keys, joins, normalization (PostgreSQL/MySQL)
- **2.3 NoSQL Databases |** Document (MongoDB), key-value (Redis), and when to reach for each over SQL
- **2.4 Writing Efficient SQL |** Aggregations, subqueries, window functions, and query optimization basics
- **2.5 Indexing & Performance |** How indexes work, query plans, avoiding N+1 queries and slow lookups
- **2.6 ORMs & Connecting Databases to Apps |** SQLAlchemy/Prisma-style ORMs, connection pooling, migrations, and environment-based DB configs



Module 3 – Data Engineering & System Design for AI (12 lessons)

- **3.1 DE Fundamentals |** Data is the fuel for AI — FDEs must be able to wrangle it
- **3.2 ETL vs ELT |** Modern: load first, transform in the warehouse (ELT wins)
- **3.3 Data Lake vs Lakehouse |** Lakehouse = Delta Lake on object storage — best of both
- **3.4 Medallion Architecture |** Bronze (raw) → Silver (cleaned) → Gold (business-ready)
- **3.5 Batch vs Streaming |** Batch for history; streaming for real-time AI
- **3.6 Data Pipelines |** Ingestion → transformation → loading → validation
- **3.7 Data Quality |** Null checks, schema validation, freshness monitoring
- **3.8 Platform Landscape |** Databricks, Snowflake, dbt, Airflow, Kafka — when to use each
- **3.9 AI System Design Fundamentals |** Requirements → components → data flow → interfaces for an AI solution
- **3.10 Designing the AI Data Layer |** Feature stores, vector stores, caches & how AI pipelines consume data
- **3.11 Scalability & Reliability Patterns |** Load balancing, queues, retries, idempotency, graceful degradation
- **3.12 System Design Case Study |** Architecting an end-to-end enterprise RAG/agent system

PART 2 - SOFTWARE ENGINEERING



Module 4 - Frontend Development for FDEs (6 lessons)

- **4.1 HTML, CSS & JavaScript Essentials** | Markup, styling, the DOM, events — the browser basics every FDE needs
- **4.2 React Fundamentals** | Components, JSX, props, state, and hooks (useState, useEffect)
- **4.3 Building UIs for AI Apps** | Chat interfaces, file upload, streaming responses, forms that talk to an LLM backend
- **4.4 State Management & API Integration** | fetch / axios, React Query, wiring a React frontend to your FastAPI/Flask backend
- **4.5 Styling & Component Libraries** | Tailwind CSS, component libraries (e.g. shadcn/ui), responsive layout for client demos
- **4.6 Lab: Build a Chat UI for an AI Assistant** | Ship a working React frontend wired to your backend API, ready to demo to a client



Module 5 - Backend & API Development (6 lessons)

- **5.1 Software Development Fundamentals** | Project structure, modularity, environments, dependencies
- **5.2 What is an API?** | REST fundamentals, HTTP methods, status codes, JSON
- **5.3 Building APIs with FastAPI / Flask** | Endpoints, request/response, validation
- **5.4 API Integrations** | Consuming third-party APIs, authentication, API keys, webhooks
- **5.5 Backend Fundamentals** | Auth, environment configs, error handling, rate limiting
- **5.6 Lab: Build & Expose a REST API** | Ship a working API that serves data to a client



Module 6 - AI-Assisted Software Engineering (8 lessons)

- **6.1 AI-First SDLC** | Prompt → Generate → Review → Refine → Ship (not: write → debug → repeat)
- **6.2 Claude Code** | Agentic CLI that reads/writes files, runs commands, fixes bugs
- **6.3 Cursor** | AI-native IDE — inline generation, codebase-aware completions
- **6.4 GitHub Copilot** | Autocomplete + chat — best for line-by-line acceleration
- **6.5 Rapid Prototyping** | 0 → working demo in 2 hours using AI tools
- **6.6 AI Code Review** | Use AI to review your own PRs before submitting
- **6.7 AI Debugging** | "Here's the error + stack trace — what's wrong?"
- **6.8 Lab: Build a Full Feature in 2 Hours Using AI Tools**

PART 3 - AI AND ANALYTICS



Module 7 - Machine Learning Basics (5 lessons)

- **7.1 What is Machine Learning?** | Supervised vs unsupervised; where ML fits for an FDE
- **7.2 Core Algorithms** | Regression, classification, clustering — the practical intuition
- **7.3 The ML Workflow** | Data → features → train → evaluate → predict (scikit-learn)
- **7.4 Model Evaluation** | Accuracy, precision/recall, overfitting, train/test split
- **7.5 Lab: Build Your First ML Model** | Train & evaluate a classifier on real data



Module 8 - Deep Learning & Transformer Architecture (5 lessons)

- **8.1 Neural Networks Explained** | Neurons, layers, weights, activation — simple mental model
- **8.2 How Networks Learn** | Forward pass, loss, backpropagation, gradient descent (intuition)
- **8.3 Key Architectures** | CNNs (images) & RNNs (sequences) — what & when, at a glance
- **8.4 Transformer Architecture** | "Attention is all you need" — the high-level mental model
- **8.5 From Deep Learning to GenAI** | Why transformers led to LLMs — the bridge



Module 9 - AI & GenAI Foundations (4 lessons)

- **9.1 Generative AI & the AI → ML → DL → GenAI evolution** | Models that generate content, not just classify
- **9.2 Tokens & Tokenization** | How LLMs see text; why tokens drive cost & limits
- **9.3 Customizing AI:** Pre-training vs Fine-tuning vs RAG · Open vs Closed models | When to use each
- **9.4 AI Application Lifecycle** | Idea → Prototype → Eval → Production → Monitor



Module 10 - Working with LLMs (4 lessons)

- **10.1 How LLMs Work & Context Windows** | Next-token prediction; the LLM's "RAM" — what fits, what gets cut
- **10.2 Hallucination** | Why LLMs make things up; mitigation strategies
- **10.3 Model Selection for FDEs** | Claude / GPT-4o / Gemini / LLaMA; the cost-speed-quality triangle
- **10.4 API vs Self-Hosted, Cost & Latency** | Token pricing, batching, caching; compliance & scale



Module 11 - Prompt Engineering Essentials (4 lessons)

- **11.1 Prompt Engineering Overview** | 80% of AI quality comes from prompts
- **11.2 Core Techniques** | Zero-shot, Few-shot, Chain-of-Thought & Prompt Chaining
- **11.3 System Prompts, Structured Output & Controls** | Persona + constraints; JSON mode; temperature
- **11.4 Evaluation & Enterprise Patterns** | A/B testing; classification, extraction, summarization, transformation

PART 3 - AI AND ANALYTICS



Module 12 – RAG & Vector Databases (5 lessons)

- **12.1 RAG Foundations** | Retrieval-Augmented Generation, embeddings & vector databases
- **12.2 Building a RAG Pipeline** | Chunking, similarity search, ingest → embed → store → retrieve → generate
- **12.3 Retrieval Strategies & Metadata Filtering** | Top-K, MMR, re-ranking; filter before vector search
- **12.4 Advanced RAG** | Self-query, HyDE, Corrective RAG, RAPTOR
- **12.5 RAG Evaluation** | Faithfulness, Answer Relevance, Context Recall (RAGAS)



Module 13 – AI Agents & Agentic AI (5 lessons)

- **13.1 What is an Agent?** | LLM + Tools + Memory + Decision Loop; tool calling via JSON schema
- **13.2 Agentic Patterns** | ReAct (Reason → Act → Observe → Repeat), planning, reflection, memory
- **13.3 Multi-Agent Systems & Frameworks** | Supervisor-worker, peer agents; LangChain & LangGraph
- **13.4 Agent Observability** | Trace every step; log tool calls; measure token usage
- **13.5 Lab:** Research Agent

Goal — build an agent that:

1. Accepts a research question
2. Searches the web · tool: web_search
3. Reads relevant pages · tool: read_url
4. Synthesizes findings into a structured report
5. Saves output to a file · tool: write_file



Module 14 – MCP & AI Workflow Automation (4 lessons)

- **14.1 What is MCP?** | Standard protocol for LLMs to call external tools safely; MCP vs REST API
- **14.2 MCP Components** | Host (Claude Desktop), Client (app), Server (tool wrapper)
- **14.3 Building an MCP Server** | Expose any tool (DB, API, file system) via MCP
- **14.4 AI Workflow Design & Automation** | Directed graphs + human checkpoints; trigger, event-driven & scheduled patterns

PART 4 - DEPLOYMENT ENGINEERING



Module 15 – DevOps & Containerization (8 lessons)

- **15.1 DevOps for FDEs** | Why DevOps matters; the build → ship → run loop
- **15.2 Git in Depth** | Branching strategies, merge vs rebase, conflict resolution, workflows
- **15.3 Docker Fundamentals** | Images, containers, Dockerfile, volumes, networking
- **15.4 Docker Compose** | Run multi-container AI apps (app + DB + vector store) together
- **15.5 Kubernetes Fundamentals** | Pods, deployments, services, scaling & self-healing
- **15.6 Kubernetes for AI Workloads** | Deploying & scaling AI services on K8s
- **15.7 CI/CD Pipelines** | GitHub Actions → test → build → deploy on every merge
- **15.8 Lab: Containerize & Orchestrate an AI App** | Docker → Compose → deploy on Kubernetes



Module 16 – Cloud Architecture & AI Deployment (15 lessons)

- **16.1 Cloud-Native Principles** | Design for failure; scale horizontally; automate everything
- **16.2 AWS Well-Architected** | 5 pillars every FDE should know before touching client cloud
- **16.3 AI Platform Deployment** | Where AI apps live: EC2, Lambda, ECS, Bedrock, SageMaker
- **16.4 Microservices & APIs** | Decompose AI apps into independently deployable services
- **16.5 Docker for AI** | Container your AI app → consistent, portable, deployable
- **16.6 Kubernetes for AI at Scale** | Orchestrate, scale & self-heal production AI services
- **16.7 Serverless for AI** | Lambda for lightweight inference; cost-effective for event-driven AI
- **16.8 Model Serving & Inference Infrastructure** | Endpoints, autoscaling, GPU vs CPU, batching for throughput
- **16.9 Deployment Strategies** | Blue-green, canary & rolling releases for AI services
- **16.10 Infrastructure as Code** | Terraform / CloudFormation – reproducible, version-controlled environments
- **16.11 CI/CD for FDEs** | Automated test → build → deploy for AI services
- **16.12 Observability & Monitoring in Production** | Logs, metrics, tracing, latency, cost & drift dashboards
- **16.13 Security Fundamentals** | IAM least-privilege; VPC isolation; secrets in AWS Secrets Manager
- **16.14 AI-Specific Security** | Prompt injection, jailbreaks, data poisoning, model & output security
- **16.15 Lab: Deploy an AI Service End-to-End** | Container → cloud → CI/CD → monitored production endpoint

PART 4 - DEPLOYMENT ENGINEERING



Module 17 - LLMops & Production Operations (10 lessons)

- **17.1 LLMops vs MLOps** | Operating LLM systems in production — what's different and why it matters
- **17.2 Model Versioning & Registry** | Track model versions; promote, roll back, and audit models (not just code)
- **17.3 Prompt & Config Management** | Version prompts, feature-flag AI behavior, ship changes safely
- **17.4 Guardrails & Safety in Production** | Input/output filtering, PII redaction, content moderation at the edge
- **17.5 Production Evaluation** | Online eval, A/B testing live traffic, human-in-the-loop feedback loops
- **17.6 Cost Management & FinOps for AI** | Token/GPU cost monitoring, budgets, rate limiting, quotas
- **17.7 Scaling & Resilience** | Request queuing, caching layers, fallback models when the primary is down
- **17.8 Environment & Release Strategy** | Dev / staging / prod separation, secrets rotation, disaster recovery
- **17.9 Monitoring, Drift & Incident Response** | Quality drift, alerting, on-call runbooks for AI systems
- **17.10 Lab: Operate a Live AI System** | Add versioning, guardrails, cost tracking & monitoring to a deployed service



Module 18 - AI Ethics, Responsible AI & Compliance (8 lessons)

- **18.1 Responsible AI Foundations** | Why ethics & safety are a business requirement, not an afterthought
- **18.2 Bias & Fairness** | Where bias comes from; detecting & mitigating it in AI systems
- **18.3 Transparency & Explainability** | Making AI decisions understandable & auditable
- **18.4 AI Regulations & the EU AI Act** | Risk tiers, obligations & what enterprises must comply with
- **18.5 Data Privacy & Compliance** | GDPR, HIPAA, data residency & handling sensitive data
- **18.6 PII & Sensitive Data Handling** | Detection, redaction, anonymization, access control
- **18.7 Governance & Audit Trails** | Policies, approvals, logging & accountability for AI in production
- **18.8 Lab: Add a Compliance & Safety Layer** | Bias check, PII redaction & audit logging on an AI service

PART 5 - CLIENT DELIVERY & BUSINESS CONSULTING



Module 19 - The FDE Role & Mindset (6 lessons)

- **19.1 What is an FDE?** | The intersection of engineering, consulting, and AI
- **19.2 FDE vs SWE** | FDEs build for clients, SWEs build products
- **19.3 The FDE Mindset** | Speed > Perfection · Empathy > Assumptions · Ship it
- **19.4 The 20/30/50 FDE** | Why 50% of the job is problem-solving & delivery, not code
- **19.5 FDE Skillset** | AI + Data + Cloud + SE + Consulting = the pillars
- **19.6 The 7-Stage FDE Playbook** | Discover → Translate → Design → Prototype → Deploy → Operate → Deliver

PART 5 - CLIENT DELIVERY & BUSINESS CONSULTING



Module 20 – Problem Discovery & Translation (8 lessons)

- **20.1 The Real Problem vs the Stated Problem** | Why clients describe symptoms, not causes
- **20.2 Stakeholder Discovery** | Who to talk to, what to ask, how to listen
- **20.3 Requirements Gathering Techniques** | Interviews, shadowing, workflow mapping
- **20.4 The Problem Translation Framework** | Business pain → technical problem statement
- **20.5 Choosing the Right Approach** | Rules vs ML vs RAG vs Agents vs "no AI needed"
- **20.6 Scoping & Success Criteria** | Define "done" and how you'll measure value
- **20.7 Risk, Data & Compliance Early** | Spotting privacy, legal & data-quality flags upfront
- **20.8 Lab: Translate a Messy Client Ask** | Turn a vague request into a one-line technical spec



Module 21 – Client Engagement & Communication (10 lessons)

- **21.1 Conducting Client Workshops** | Facilitate, align stakeholders, extract requirements
- **21.2 Communicating to Technical & Non-Technical Stakeholders** | Same solution, two languages
- **21.3 Technical Storytelling** | Problem → journey → outcome; make people care
- **21.4 Solution Presentation & Demos** | Show, don't tell; run a demo that sells
- **21.5 Managing Expectations** | Scope, timelines, and saying no gracefully
- **21.6 Handling Difficult Stakeholders** | Skeptics, scope-creep, and conflicting priorities
- **21.7 Writing Technical Briefs & Proposals** | Clear, persuasive, approvable documents
- **21.8 Implementation Roadmaps** | Phasing, milestones, and quick wins
- **21.9 Running the Feedback Loop** | Iterate in front of the client without losing control
- **21.10 Lab: Pitch & Present a Solution** | Present a POC to a mock client panel



Module 22 – Enterprise Delivery, Pricing & Consulting (14 lessons)

- **22.1 The FDE Delivery Model** | How FDEs embed, deliver, and create value
- **22.2 Converting Business Problems into AI Solutions** | End-to-end, in practice
- **22.3 Designing Solutions Around Business Objectives** | Tie every build to measurable value
- **22.4 Scoping an Engagement** | Breaking a project into phases, effort & deliverables
- **22.5 Cost Estimation & Pricing a Project** | Estimating effort, AI running costs & pricing the engagement
- **22.6 Building Rapid POCs** | Prove value fast, de-risk the project
- **22.7 Enterprise Project Planning** | Milestones, risks, realistic timelines
- **22.8 Delivery Best Practices** | Quality, documentation, and handover
- **22.9 Training & Enabling the Client** | Make the client independent after you leave
- **22.10 Measuring & Reporting ROI** | Show the business impact you delivered
- **22.11 Real-World Client Engagement Scenarios** | Case-based practice
- **22.12 Portfolio Building** | Case studies & GitHub projects that prove you can deliver
- **22.13 LinkedIn & Personal Branding** | Get found and hired as an FDE
- **22.14 The End-to-End Enterprise AI Delivery Framework** | The full FDE method, start to finish

HANDS-ON PROJECTS - REAL-WORLD END-TO-END BUILDS

Ten full, guided end-to-end projects – **8 AI Track** projects and 2 **ML Track** projects, across different industries. Each is a real enterprise use case taken all the way through the FDE playbook – discover, build, deploy, and deliver a live solution.

AI Track (8 projects)

Project 1 Enterprise Knowledge Assistant (**Healthcare / Banking**)

A hospital / bank support desk is drowning in repetitive questions. Build an AI assistant that answers accurately from the client's own documents, with sources and safe fallbacks.



P1.1 Discover & Translate

Scope the FAQ problem; write the technical spec

P1.2 Build the RAG Pipeline

Ingest PDFs → chunk → embed → vector store → retrieve

P1.3 Add Guardrails & Grounding

Source citations, "I don't know" fallback, PII filtering

P1.4 Deploy & Expose

Containerize, put behind an API, ship a simple chat UI

P1.5 Evaluate & Deliver

RAGAS accuracy check, cost tracking, handover

Project 2 Document Processing & Automation Agent (**Insurance / Operations**)

An operations team manually reads incoming documents (invoices, claims, contracts), extracts data, and files it. Build an agent that reads documents, extracts structured data, validates it, and pushes it into a system automatically.



P2.1 Discover & Translate

Map the manual workflow; define what to automate

P2.2 Build the Extraction Agent

Read documents → extract structured JSON → validate

P2.3 Tool Calling & Integration

Push validated data to a database / API via tools

P2.4 Human-in-the-Loop & Automation

Approval checkpoints; trigger/scheduled runs

P2.5 Deploy, Monitor & Deliver

Ship it, add observability & cost controls, hand over

Project 3 Customer Support Copilot (**SaaS / Technology**)

A SaaS company's support team is overwhelmed by tickets that repeat the same 20 issues. Build an AI copilot that drafts replies from the knowledge base and product docs, with a human approving before send.



P3.1 Discover & Translate

Shadow the support team; scope what to automate vs assist

P3.2 Build the Knowledge Layer

Ingest docs, past tickets & macros into a searchable store

P3.3 Build the Frontend

React-based agent dashboard - suggested reply, edit, approve/send

P3.4 Wire Frontend to Backend

FastAPI backend serving suggestions to the React UI in real time

P3.5 Deploy & Deliver [Container]

Container → cloud → CI/CD → live pilot with a support team

Project 4 Sales & Operations Analytics Dashboard (**General Business / Sales**)

Leadership gets a static monthly spreadsheet and wants to ask questions of their data in plain English. Build a data pipeline plus an AI-powered dashboard that answers natural-language questions over live business data.



P4.1 Discover & Translate

Identify the real decisions leadership needs the data for

P4.2 Build the Data Pipeline

ETL/ELT from source systems into a clean, queryable warehouse

P4.3 Build the Analytics Backend

Natural-language-to-SQL layer + API over the warehouse

P4.4 Build the Dashboard Frontend

React charts/tables, chat-style query box, drill-downs

P4.5 Deploy, Monitor & Deliver

Ship to cloud, add usage monitoring & cost tracking, handover

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AI Track (8 projects)

Project 5 Multi-Agent Workflow Automation Platform (Logistics / Insurance)

A logistics or insurance client has a multi-step process (intake → validation → routing → approval) currently done manually across several tools. Build a multi-agent system that automates the full workflow end-to-end.

P5.1 Discover & Translate

Map the multi-step process; decide what each agent owns

P5.2 Design the Multi-Agent System

Supervisor + worker agents, tool access, handoff logic

P5.3 Build & Integrate

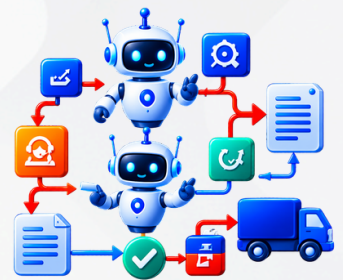
Connect agents to real APIs/DBs; add human-approval checkpoints

P5.4 Harden for Production

Guardrails, cost limits, observability, fallback on agent failure

P5.5 Deploy & Deliver

Container → cloud → CI/CD → monitored production rollout, handover



Project 6 AI Legal Contract Analyzer & Negotiation Assistant (Legal)

A law firm or corporate legal team spends hours manually reviewing contracts for risky clauses. Build an advanced AI system that reads contracts, flags risk, compares against a standard playbook, and suggests redline language.

P6.1 Discover & Translate

Work with lawyers to define what "risky clause" means; get access to the firm's contract playbook

P6.2 Build the Legal Knowledge Base

Ingest playbooks, past contracts & precedent clauses into a specialized retrieval system

P6.3 Build the Advanced Reasoning Pipeline

Multi-step AI reasoning: extract clauses → classify risk → compare to playbook → draft redlines

P6.4 Add Guardrails for High-Stakes Use

Human-in-the-loop review for every suggestion; full citation of source clauses; confidence scoring

P6.5 Deploy & Deliver

Ship as a secure internal tool, train the legal team, prove time saved per contract reviewed



Project 7 Autonomous Financial Research & Investment Analyst Agent (Finance / Investment)

An investment team spends days manually researching a company before making a decision. Build a multi-agent system that autonomously researches a company – financials, news, competitors – and produces an investment memo.

P7.1 Discover & Translate

Understand the analyst's real research process and what a "good" memo looks like

P7.2 Design the Multi-Agent Research System

Separate agents for financial data, news sentiment, competitor analysis & report writing

P7.3 Build Tool-Calling & Data Integration

Connect agents to financial data APIs, news feeds & web search; ground every claim in a real source

P7.4 Add Fact-Checking & Guardrails

Cross-check agent outputs against each other; flag low-confidence claims; prevent hallucinated numbers

P7.5 Deploy & Deliver

Ship as an on-demand research tool, prove accuracy against analyst-written memos, hand over with full source trails



Project 8 Multimodal Medical Diagnostic Assistant (Healthcare)

A radiology or diagnostics team needs to review medical images alongside patient notes to flag potential issues faster. Build a multimodal AI assistant that reads both images and clinical text together to support (not replace) a doctor's diagnosis.

P8.1 Discover & Translate

Work with clinicians to define scope, required accuracy, and where a human must always stay in the loop

P8.2 Build the Multimodal Data Pipeline

Organize and pair medical images with clinical notes and patient history securely

P8.3 Build the Multimodal AI Pipeline

Combine a vision model (image analysis) with an LLM (clinical notes) to generate a combined assessment

P8.4 Add Safety, Compliance & Human Review

Mandatory clinician sign-off on every output; full audit logging; HIPAA-aligned data handling

P8.5 Deploy & Deliver

Ship as a clinician-facing tool, validate against real diagnostic outcomes, prove it as a safe second opinion – not a replacement



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ML Track (2 projects)

Project 9 Real-Time Fraud Detection System (Banking / Fintech)

A bank or payments company loses money every day to fraudulent transactions that slip past simple rule-based checks. Build a machine learning system that scores every transaction for fraud risk in real time, at the speed money actually moves.

P9.1 Discover & Translate

Understand the bank's current rule-based system and its blind spots; define acceptable false-positive rates

P9.2 Build the Data Pipeline

Ingest transaction streams, account history & device/location signals; engineer features under strict time constraints

P9.3 Train the Fraud Model

Handle extreme class imbalance (fraud is rare); train, validate & tune a classification model (XGBoost / gradient boosting)

P9.4 Real-Time Scoring & Explainability

Serve predictions in milliseconds; make every flagged transaction explainable for compliance review

P9.5 Deploy, Monitor & Deliver

Ship as a low-latency API, monitor for model drift as fraud patterns evolve, hand over with a full audit trail



Project 10 Predictive Maintenance for Industrial Equipment (Manufacturing)

A factory's machines break down unexpectedly, causing costly downtime. Build a system that predicts equipment failure before it happens, using sensor data, so maintenance can be scheduled proactively instead of reactively.

P10.1 Discover & Translate

Understand what "failure" means to this client; identify which sensors and machines matter most

P10.2 Build the Sensor Data Pipeline

Ingest streaming time-series data (vibration, temperature, pressure); clean and resample noisy sensor readings

P10.3 Train the Prediction Model

Build a time-series model that forecasts remaining useful life / failure probability per machine

P10.4 Build the Alerting Dashboard

React dashboard showing machine health, risk scores, and recommended maintenance windows

P10.5 Deploy, Monitor & Deliver

Ship to the factory floor's environment, monitor prediction accuracy over time, prove downtime reduction



Bangalore



Bhubaneswar



Dubai



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